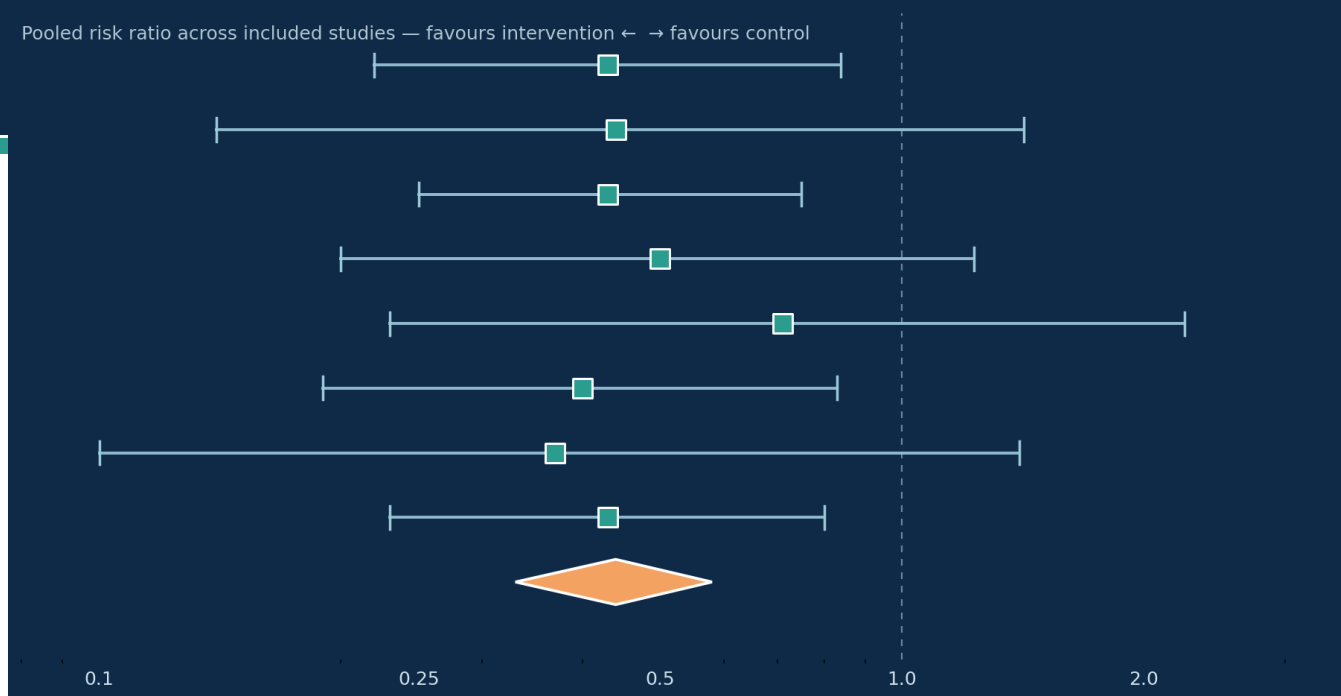


Clinical Research Assistant

An AI co-investigator for systematic reviews, meta-analyses, and evidence synthesis — built for hospital research offices, academic medical centres, and clinical research organisations.



Sample forest plot output — every effect estimate traces back to a real database hit.

Why this exists

A high-quality systematic review takes a small team **6–12 months**: scoping the question, building search strategies, screening thousands of abstracts, extracting data, running the statistics, assessing bias, and writing it up. Most of that work is repetitive, methodology-bound, and easy to get wrong in ways that don't show up until a reviewer rejects the manuscript.

The **Clinical Research Assistant** compresses the methodological scaffolding into a guided, auditable workflow — without ever fabricating evidence. Your investigators stay in the driver's seat for clinical judgment. The assistant handles the parts that should never have been manual in the first place.

What the assistant does

Five guided evidence workflows, a continuous-monitoring service, a local research store that grows in value the more you use it, and a regulatory-grade **electronic data-capture (eCRF/EDC)** subsystem for running your own studies. Each workflow launches from natural language or a slash command and produces a structured, citation-anchored output you can take into a manuscript, a registration, or a grant application.

1 Meta-analysis workflow

End-to-end: from a research question to a forest plot. The assistant walks the user through **PICO** confirmation, runs the literature search across every database your institution has licensed, presents candidate studies for inclusion/exclusion, captures the extraction table, then computes the pooled effect estimate with heterogeneity statistics and renders a publication-grade forest plot inside an isolated compute sandbox.

***Outcome** — what used to be a multi-week analyst engagement is a single guided session. Every PMID in the output traces back to a real database hit — the model is structurally prevented from inventing citations.*

2 Search-strategy builder

Constructs Boolean queries with MeSH terms, field tags, and database-specific syntax across **PubMed, Europe PMC, Embase, Cochrane Library, Scopus, and Web of Science** — whichever your institution has licensed, in a single fan-out. Iterates on **broaden / tighten** suggestions with the user until the strategy is registration-ready.

***Outcome** — information specialists get a defensible, multi-database, reproducible search string in minutes instead of half a day, with the iteration history preserved.*

3 Systematic-review protocol drafter

Generates a **PRISMA-P-aligned** protocol skeleton — background, objectives, eligibility, search methods, screening plan, data items, risk-of-bias plan, synthesis approach — ready to deposit in **PROSPERO**.

***Outcome** — a protocol-ready first draft your methodologist edits, not writes from scratch. Helps surface gaps before they cost you a peer-review round.*

4 Risk-of-Bias assessor

Supports the major instruments: **RoB 2** (RCTs), **ROBINS-I** (non-randomised), **Newcastle-Ottawa** (observational), **QUADAS-2** (diagnostic accuracy). Walks domain-by-domain, asks the operator the judgement questions, and produces the structured assessment table for the review.

***Outcome** — consistency across reviewers and a clean audit trail of why each judgement landed where it did.*

5 General clinical Q&A

For the questions that aren't a full systematic review: background reading, definition of methods, navigation of guidelines. The assistant uses web and Wikipedia tools, but is **hard-prevented from quoting effect sizes, PMIDs, or guideline citations from training data** — a regex-level validator blocks unsupported clinical claims before they reach the user.

***Outcome** — safe-by-default exploratory Q&A that won't seed a manuscript with hallucinated evidence.*

6 Living-review watches (scheduled monitoring)

A meta-analysis or systematic review doesn't have to be a snapshot. Pin a PICO and a search strategy as a **watch**, set a cadence (daily / weekly / monthly), and the assistant re-runs the search on schedule, diffs against the baseline corpus, triages newly published papers against the original PICO, and **notifies the team when something materially shifts the evidence base**.

***Outcome** — living reviews stop drifting out of date. Guideline committees and HTA bodies get alerted to practice-changing evidence as it lands.*

7 Electronic data capture (eCRF / EDC)

Beyond synthesising other people's evidence, the assistant runs **your own studies' data collection** — an electronic case-report-form (eCRF) and data-capture (EDC) subsystem built to **CDISC** and **21 CFR Part 11 / ALCOA+** conventions. Paste a protocol and the assistant drafts CDASH-aligned forms plus a visit schedule (a designer reviews — never auto-publish); versioned, immutable definitions export as **CDISC ODM-XML**. Two capture surfaces — a **site EDC** screen and a consent-gated, magic-link **participant ePRO** surface — validate on entry, with hard/soft edit-checks feeding a full query workflow. Subject PHI lives in a **separate clinical-data store**, never sent to the model. E-signatures bind to the exact signed data, lock against edits, and roll up to a subject-casebook sign-off; source-data verification and a **tamper-proof, append-only audit trail enforced by the database** capture every change with who, when, old→new value, and reason.

***Outcome** — the weeks-long investigator↔data-manager round-trip at study start-up collapses to a guided session, and the resulting capture system is audit-ready by construction.*

What makes it different

Differentiator	Why it matters
Anti-hallucination enforced in code, not just prompt	A regex validator and tool-gated PMID rule block fabricated citations even if the underlying model drifts. Your investigators can trust the bibliography.
Source-agnostic search across paid + open databases	PubMed, Europe PMC, Embase, Cochrane Library, Scopus, Web of Science — all live behind a pluggable PaperSource interface and fanned out in parallel with cross-source de-duplication. One admin panel for credentials, rate-limits, enable/disable.
Local research cache + RAG over pulled content	Every abstract, full-text article, MeSH lookup, and extraction table you pull lands in the local research store. A retrieval-augmented pipeline searches that store first, so the same paper isn't re-fetched — or re-billed — across reviews. Pull paid content once; reuse it across the entire research programme.
Sandboxed statistical compute	All Python analysis runs inside a Docker container with networking disabled, read-only inputs, and capped CPU/memory. The model can run a random-effects meta-analysis without ever touching the host or the wider internet.
Workflow-gated tools	The assistant cannot skip ahead. It can't run a meta-analysis before a PICO is confirmed; it can't fetch full text before studies are selected. The methodology <i>is</i> the guardrail.

Regulatory-grade data capture built in	The eCRF/EDC subsystem follows CDISC (ODM-XML export, CDASH naming) and 21 CFR Part 11 / ALCOA+ : subject PHI in a separate store, e-signatures bound to the signed data, source-data verification, and an append-only audit trail enforced by the database — not just application code. Author a CRF, deploy it, and collect site + participant data without leaving the platform.
Flexible, secure deployment	Deploy on-premises or in your cloud. OAuth2 / OpenID Connect integrates with your existing identity provider; role-based access control scopes each workflow per user; sensitive material (API keys, PHI) lives in your secrets manager , never in plaintext config.
Full conversation persistence	Every turn — PICO, included studies, extraction table, generated code, plot — is stored. Reproducing an analysis a year later means re-opening the thread.

What it costs to run

Two variable costs to plan around: LLM tokens on AWS Bedrock and web-search calls on Tavily. Everything else — paper database access, paper storage, the FastAPI app, the sandbox — sits on infrastructure you already own. Paid bibliographic databases (Embase, Scopus, Cochrane, etc.) are assumed to be covered by your existing institutional subscriptions.

ACTIVE RESEARCH DAY (PER USER)

**\$200 –
\$300**

Covers Bedrock tokens + Tavily for a productive day: PICO refinement, multi-database search, screening + extraction across dozens of papers, statistical analysis with code generation, forest-plot rendering.

RE-OPEN / INSPECT A THREAD

≈ \$0

Past tool results (papers, extractions, generated code, plots) are already in the local store. Re-opening costs only the few tokens of the current turn.

RE-USE A PAPER ACROSS REVIEWS

≈ \$0

The RAG pipeline hits the local research store first. The same landmark trial reused across cardio / onco / ID reviews is paid for once.

Cost-control levers built in

Cost-control lever	How it keeps spend predictable
Local cache + RAG	Re-use of pulled abstracts and full text across reviews — no repeat API spend on content you've already paid for.
Workflow-gated tools	Expensive tools (sandbox, full-text fetch) are unreachable until the workflow stage needs them — no idle calls.
Per-turn ceilings	max_model_requests and agent_timeout_seconds bound runaway loops to a known maximum cost.
Configurable model tier per workflow	Drop to a cheaper Sonnet / Haiku variant when full Opus reasoning isn't required.
Tavily scoped to two flows	Tavily is consumed only by general Q&A + protocol drafting; the clinical search runs against bibliographic databases and does not consume Tavily credits.

A token-heavy day still lands comfortably **below the cost of an equivalent human-analyst day** — and per-paper marginal cost flattens as your local store grows.

Deployment & governance

Area	What you get
Flexible deployment	Runs on-premises or in your cloud (AWS, GCP, Azure). Single-tenant by default; same codebase, your choice of trust boundary.
Authentication	OAuth2 / OpenID Connect — drops into Okta, Azure AD, Auth0, Keycloak, or your homegrown IdP. No bespoke user database.
Authorisation	Role-based access control per user. New-researcher onboarding is a one-step provisioning action that scopes their search-API entitlements, model access, and visibility into shared review threads.
Sensitive data	API keys, Bedrock credentials, and any PHI live in a secrets manager (AWS Secrets Manager, HashiCorp Vault, etc.) — never in plaintext config or SQLite.
Data egress	Outbound traffic is limited to your configured bibliographic APIs and your Bedrock endpoint. No telemetry. No shared multi-tenant cloud — LLM calls go to your own AWS account, so PHI-grade compute stays inside your trust boundary.
Audit	Every turn is persisted with full message + tool-call history. Re-opening a thread reproduces the analysis exactly. Exportable as part of a regulatory submission package.

On the roadmap

Beyond the v1 capability set above, the architecture was built to absorb these without rebuilding the core. Each is a tractable next increment, not a moonshot.

Source-document extraction

The eCRF / EDC subsystem (§7 above) now ships. The remaining piece of automated data collection is **source-document extraction** — point the assistant at structured exports from your EHR / registry / trial-management system and have it populate the extraction table (or pre-fill eCRF instances) for retrospective studies or patient-level meta-analyses, with an audit trail of which source row produced which output cell.

Deepening eCRF compliance toward a formal validation pack

The eCRF subsystem is built to Part 11 / ALCOA+ conventions; the next increments formalise it for an audited deployment: full password re-authentication at signing, study-level (not just subject-level) lock, and a documented computer-system-validation (IQ/OQ/PQ) package.

Patient-facing research handouts

Generate **plain-language summaries** of a study's objectives, what participation involves, and what the early evidence suggests — at a configurable reading level, in the patient's preferred language. Designed for **research participant recruitment**, **shared-decision-making conversations**, and **post-study return-of-results** obligations under modern IRB / ethics guidance. Sources back to the same evidence base the clinical workflow uses, so the lay summary and the manuscript are in lock-step.

Research-gap analysis specialist

Given a body of literature, surface where the evidence base is thin — by population, intervention, outcome, geography, or study design — so investigators can prioritise the next grant proposal where the field actually needs new data.

Group-level living-review subscriptions

Group watches with quorum-based notification rules — designed for guideline committees and HTA bodies who need consensus signalling on practice-changing evidence rather than per-user alerts.

In short

If your researchers spend more time finding and formatting evidence than interpreting it, this is what you point them at. They keep the clinical judgement. The assistant absorbs the methodological choreography — and writes it down as it goes.

Appendix A — Capability Matrix

Workflow	Trigger / slash	Inputs collected	Tools available	Structured outputs	Hardened guardrails
Meta-analyses	"meta-analysis on...", "pooled effect of...", /meta	PICO → included studies → extraction table	search_papers, rag_search, mesh_lookup, fetch_pmc_fulltext, sandbox_exec, calculator	PICO card, study-selection card, extraction card, forest plot + summary	PMIDs must come from search; plot rules enforced in sandbox
Search strategy	"build a search strategy...", /search	Concept terms, MeSH, filters	mesh_lookup, search_papers (preview counts)	Boolean string per database, MeSH map, broaden/tighten suggestions	None needed — output is the query itself
SR protocol	"draft a protocol...", /protocol, /prisma	PICO + scope choices	web_search, wikipedia, fetch_document	PRISMA-P-shaped protocol sections	No effect-size or PMID synthesis allowed
Risk of Bias	"risk of bias...", "RoB 2", /rob	Tool choice + per-study source	fetch_pmc_fulltext, rag_search, read_file	Domain-by-domain judgements with rationale	Operator must confirm each judgement
General Q&A	Everything else, /general	Free-form question	web_search, wikipedia, fetch_document, rag_search, read_file, describe_image	Cited prose answer	Regex validator blocks unsupported clinical claims
Watch triage (background)	Configured per-PICO schedule	New PMIDs since last run	(none — text-only)	Per-paper triage + run summary + optional notification	Same anti-hallucination posture as user-facing flows
eCRF authoring	Form Builder UI · "draft CRFs from protocol"	Protocol text → form definitions	ecrf_design specialist	Versioned form definitions, ODM-XML export	Integrity-validated definitions; human review before publish; AI never auto-publishes
EDC capture (site)	Data Capture UI	Subject data against deployed forms	(collection API)	Captured item data, queries, signatures	Edit-checks (hard block / soft query); signed forms edit-locked; append-only audit (DB-enforced)
ePRO capture (participant)	Magic-link /epro	Patient-reported outcomes	(token-scoped API)	Captured item data	Consent gate; per-subject token scope; same edit-checks + audit

Tool inventory

Category	Tool	What it does
Clinical	search_papers	Fan-out search across enabled databases (PubMed, Europe PMC, Embase, Cochrane, Scopus, Web of Science); dedupes by PMID → DOI → source-id
Clinical	rag_search	Retrieval-augmented search over the local research store — checks pulled content before re-issuing API calls
Clinical	mesh_lookup	MeSH term resolution + tree-walk
Clinical	fetch_pmc_fulltext	Full-text retrieval; results cached into the local research store
Data science	sandbox_exec	Docker-isolated Python with pandas / numpy / scipy / statsmodels / matplotlib / seaborn / forestplot — no network, capped CPU & memory
Data science	python_repl	Lightweight in-process Python for arithmetic-only ops
Data science	calculator	Safe arithmetic evaluator
General	web_search	Tavily-backed search for non-clinical context
General	wikipedia	Definitional / background lookups

Category	Tool	What it does
General	fetch_document	Fetch + extract text from arbitrary URLs
General	read_file	Read user-uploaded documents (PDF, DOCX, TXT)
General	describe_image	Vision model for chart / figure / scan interpretation

Configuration surface

Setting	Where	Why an admin cares
Paper-source enable / disable, rate limits, credentials	Admin panel (credentials read from your secrets manager)	Plug in your institution's PubMed / NCBI key, Embase, Cochrane, Scopus, Web of Science subscriptions — toggle without redeploying
Identity provider	OIDC client config	Wire to Okta / Azure AD / Auth0 / Keycloak / homegrown
Role-based access	Admin panel	Per-user scoping of search entitlements, model tier, and shared-thread visibility
LLM model + region	Per-workflow config	Pin a Sonnet / Opus / Haiku version your governance team has approved; drop to cheaper tier for lighter workflows
Sandbox limits	Config (sandbox_timeout_seconds, sandbox_memory_mb, sandbox_cpu)	Cap compute per-analysis
Context window & summarisation	Config (context_window_messages, summarize_after_messages)	Keep long meta-analysis threads from blowing past model limits
Runaway-loop bounds	Config (max_model_requests, agent_timeout_seconds)	Hard ceilings on per-turn cost

Appendix B — Compliance & guardrails at a glance

- **No fabricated citations.** Clinical PMIDs are only emitted when they originated from a live `search_papers` call. The general Q&A flow has a separate regex validator that rejects answers attempting to quote effect sizes, PMIDs, or guideline statements not present in the conversation's tool results.
- **No silent skipping.** Workflow specialists hide downstream tools until the upstream stage is confirmed by the user — e.g. `sandbox_exec` is invisible to the model until an extraction table exists.
- **No host access from generated code.** The Python sandbox runs in a Docker container with networking disabled, the script and inputs mounted read-only, and only a designated output directory writable.
- **No plaintext secrets.** API keys, model credentials, and any PHI live in your secrets manager and are read at runtime — never persisted in config files or the local database.
- **No third-party LLM intermediary.** Model calls go directly to your AWS Bedrock account; you choose the region; the assistant has no shared multi-tenant cloud.
- **Full reproducibility.** Threads persist every user message, model message, tool call, tool result, and structured output. The same thread re-opened tomorrow reproduces today's forest plot.